Unified Contact Model and Hybrid Motion/force Control for Teleoperated Manipulation in Unknown Environments

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***Abstract*—Teleoperated manipulation under the human intelligence is an effective solution to confront complicated tasks in unknown environments. However, the uncertain contact in environments is the main challenge to achieve good teleoperation, and some inevitable issues such as nonlinearities, various uncertainties, constraints, and communication delays in local and remote robots should also be taken into account. In this article, a unified contact model is proposed to be the targeted environment interaction for remote robot, which can cover various conditions such as free motion and rigid contact by setting different model parameters. Subsequently, a hybrid motion/force controller is developed to cope with nonlinearities and various uncertainties by adaptive robust technique, thus guaranteeing system stability and good transient convergence to the unified contact model. Particularly, to handle misteleoperation or sudden change of conditions described in unified contact model, a model predictive control-based contact optimization method is developed to be the outer loop of hybrid motion/force controller, which plans the desired motion and force trajectories that meet the state and targeted interaction constraints. By the estimation and transmission of environment parameter, the environment dynamics is reconstructed in the local side, which provides the effective force feedback of remote contact environment for human operator. Since the signal transmitted is replaced by the estimated environment parameter, the power-cycle in communication channel is eliminated. This design can avoid the passivity stability problem caused by communication delays, so the stability and good transparency under delays can be guaranteed by the separate design of local and remote controllers. The comparative experiments are implemented and verify the effectiveness of proposed framework for teleoperated manipulation.**

***Index Terms*—Contact optimization, hybrid motion/force control, teleoperated manipulation, unified contact model, unknown environments.**

I. Introduction

With the progress of automation and robotics, the human–robot–environment interacted teleoperation provides a more secure and efficient solution to implement the tasks in the remote and hazardous environment [1], [2], [3]. A bilateral teleoperation system mainly consists of the human operator, local robot, communication channel, remote robot, and environment [4]. In a teleoperated task, the human operator plays the role of decision maker and provides the action guidance via local robot (e.g., haptic device) as well as the error correction of remote robot in a safe indoor place. Meanwhile, the remote robot executes the tasks in environment and provides the force feedback for the human operator [5].

For some typical manipulation tasks (e.g., industrial processing, rescue, surface scanning in medical imaging, etc.), the precise and secure contact force between the remote robot and environment is of great significance [6]. If the contact force is not well applied, the objects or environment may be damaged or even destroyed, so the teleoperation of these tasks puts forward the higher requirements for human operator, which may let the human operator undergo a heavy psychological strain and fail to effectively guide the remote robot to the desired position. Although some training systems are developed to train the human operator for these typical manipulation tasks [7], the human operator may take lots of time and energies to obtain the well-trained abilities. Besides, these training systems are costly and only available for one specific task. Since the teleoperated manipulation may encounter various conditions, such as free motion and different rigid contacts, which has the complicated dynamic characteristics in motion, force, and impendence, so it is hard for human operator to handle all these conditions with the only force feedback.

To obtain the precise and secure contact force without the concentrated and laborious operation of human operator, some research works are carried out on the force or impedance control

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of teleoperation system [8]. Sun et al. [9] proposed a sliding mode control with external force regulation, which is to identify the different motion situations of remote robots. Lu et al. [10] proposed a fuzzy-observer-based hybrid force/position control to achieve the position tracking in soft-handling environment. However, there are still limited research works that have been developed in this area, and some issues are still challenging [11], [12], for example, how to apply the exact control laws that allows the remote robot to adapt to the various conditions, such as free motion and rigid contact, while obtain the compliance during the uncertain contact to reduce the workload of human operator. Besides, it is also worthwhile to investigate how to provide the accurate force feedback that make the human operator be aware of the current condition of remote robot, which help improve his/her guiding ability.

Apart from the uncertain force contact in teleoperated manipulation, achieving the stability and transparency of teleoperation system under communication delays is also a significant issue [13]. The passivity control theory based teleoperation control framework (e.g., wave variables [14], time-domain passivity method [15], etc.) are widely used currently to guarantee the system's stability under communication delays. However, the above-mentioned frameworks may cause the signals' distortion so that the position command and force feedback may be inaccurate, which deteriorate the system's transparency [16]. In order to improve the transparency, several modifications (e.g., linear matrix inequality optimization of delayed communication channel [17], adaptive fuzzy observer based signals' distortion compensation [18], etc.) are proposed, but these designs still exist the power-cycle formed by the forward position and feedback force signals, which may not completely eliminate the signals' distortion. A framework with environment parameter transmission is proposed to replace the force signals in the communication channel [13], [19], which avoids the power-cycle thus guaranteeing the system's stability, but the environment considered is simply known with static interaction. Besides, the nonlinearities and various uncertainties (e.g., parametric uncertainties, uncertain nonlinearities, etc.) of local and remote robots will also affect the transparency of teleoperation system. Several advanced control methods (e.g., adaptive control [20], fuzzy logics [18], neural networks [21], etc.) are proposed to cope with nonlinearities and various uncertainties, which allows the teleoperated robots to implement the more complicated tasks.

Based on the above-mentioned analysis, most of the existing works study the transparency improvement of teleoperation system, but there are still limited research works that focus on the contact of remote robot in the unknown environments. Therefore, a novel teleoperation control framework is proposed in this article, where the unified contact model and hybrid motion/force control are developed to extend the system that can be used in manipulation with uncertain contact, which can reduce the workload and improve the ability to guide the manipulation of human operator. Besides, the inevitable issues such as nonlinearities, various uncertainties, constraints, and communication delays are still considered. The main contributions are as follows.

1) A unified contact model is proposed to be the targeted environment interaction for remote robot. By setting different model parameters, it covers various conditions (i.e., free motion, rigid contact) that occurs during teleoperation in one equation.
2) An adaptive-robust-based hybrid motion/force controller is designed for remote robot to cope with nonlinearities and various uncertainties. It guarantees the system to be stable and transiently convergent to the unified contact model, so that the compliant manipulation under uncertain contact is achieved.
3) To handle the misteleoperation and sudden change of conditions described in unified contact model, a state-triggered contact planner is designed. It is as the outer loop of hybrid motion/force controller, which plans the desired motion and force trajectories satisfying the state constraints of remote robot and targeted interaction constraints of unified contact model.
4) By transmitting the estimated environment parameter, the environment dynamics is reconstructed in the local side and served as force feedback. As a result, the human operator can accurately sense the compliant contact force guaranteed by hybrid motion/force controller.
5) Since the signal transmitted in the communication channel is replaced by the estimated environment parameter (nonpower signal), the power-cycle (formed by the forward position and feedback force signals) in the traditional teleoperation control is eliminated. As a result, it directly avoids the passivity stability problem caused by communication delays, and the stability and good transparency under delays can be guaranteed by the separate design of local and remote controllers.

II. Preliminaries

A. Modeling of Local and Remote Robots

The dynamics of local and remote robots in Cartesian space can be described as follows:

$$M_l(q_l)\ddot{X}_l + C_l(q_l,\dot{q}_l)\dot{X}_l + G_l(q_l) + d_l = u_l + F_h \quad (1)$$

$$M_r(q_r)\ddot{X}_r + C_r(q_r,\dot{q}_r)\dot{X}_r + G_r(q_r) + d_r = u_r + F_e \quad (2)$$

where $i = l, r$ represents the local robot and the remote robot with ξ degrees of freedom (Dof), respectively. $q_i, \dot{q}_i, \ddot{q}_i \in \mathbb{R}^{\xi\times 1}$ are the joint angle, velocity, and acceleration in joint space. Define ρ as the controlled Dof in Cartesian space. $X_i \in \mathbb{R}^{\rho\times 1}$ is the position of the end effector. Note that q_i and X_i satisfy the relationship as $J_i(q_i) = \partial X_i(q_i)/\partial q_i$, which is the Jacobian matrix. $M_i(q_i) \in \mathbb{R}^{\rho\times\rho}$ is the mass inertia matrix, $C_i(q_i,\dot{q}_i) \in \mathbb{R}^{\rho\times\rho}$ is the coriolis/centrifugal force matrix, $G_i(q_i) \in \mathbb{R}^{\rho\times 1}$ is the gravitational force matrix, $d_i \in \mathbb{R}^{\rho\times 1}$ is the external disturbances, $u_i \in \mathbb{R}^{\rho\times 1}$ is the control input, $F_h \in \mathbb{R}^{\rho\times 1}$ is the operating force exerting on the local robot, $F_e \in \mathbb{R}^{\rho\times 1}$ is the contact force exerting on the environment, both F_h and F_e can be measured by the force sensor. Some properties and assumptions of (1) and (2) are as follows:

Property 1: M_i is the symmetric positive-definite matrix defined as follows:

$$\eta_{i1} I_{\rho\times\rho} \le M_i \le \eta_{i2} I_{\rho\times\rho} \quad (3)$$

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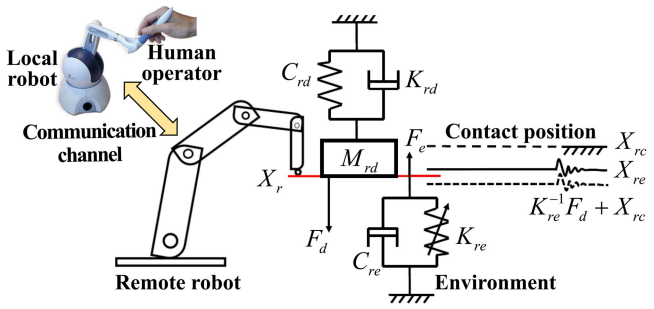


Fig. 1. Bilateral teleoperation system with manipulation tasks.

where η_{i1} and η_{i2} are the positive scalars.

Property 2: $M_i - 2C_i$ is the skew-symmetric matrix.

Property 3: By the proper selection of parameters, part of dynamics (1) and (2) is linear and can be written as follows:

$$M_i(q_i)\ddot{X}_{id} + C_i(q_i, \dot{q}_i)\dot{X}_{id} + G_i(q_i) = \varphi_i^T(q_i, \dot{q}_i, \dot{X}_{id}, \ddot{X}_{id})\theta_i + \Delta_i \tag{4}$$

where φ_i is the vector of known smooth functions, θ_i is the unknown parameter of the local or remote robot, $\dot{X}_{id}$ and $\ddot{X}_{id}$ are the desired velocity and acceleration signals, Δ_i is the modeling error.

Assumption 1: The bounds of the parametric uncertainties and uncertain nonlinearities are known and described as follows:

$$\theta_i \in \Omega_{\theta_i} \triangleq \{\theta_i : \theta_{i,\min} \le \theta_i \le \theta_{i,\max}\} \tag{5}$$

$$\tilde{d}_i \in \Omega_{\tilde{d}_i} \triangleq \{\tilde{d}_i : \| \tilde{d}_i \| \le \delta_i(X_i, \dot{X}_i, t)\} \tag{6}$$

where $\tilde{d}_i = \Delta_i + d_i$ is the lumped effect of the modeling error and external disturbances.

B. Modeling of Environment Dynamics

Considering the manipulation tasks in the teleoperation process, it is important to obtain the accurate environment dynamics when the contact occurs. Assume that the manipulation tasks have the stable and single contact, the unknown environment dynamics (including the unknown contact position and parameters) can be described as the Kelvin–Voigt model, which is a linear damper–spring system as follows:

$$K_{re}(X_r - X_{rc}) + C_{re}\dot{X}_r = F_e \tag{7}$$

where K_{re} and C_{re} are the unknown stiffness and damping of the environment, respectively, X_{rc} is the contact position of the remote robot and environment.

C. Problem Formulation

For a typical teleoperation system with manipulation tasks, as shown in Fig. 1, considering the uncertain contact in unknown environment, as well as the nonlinearities, various uncertainties and constraints, bounded F_h, stable and single contact force F_e, and communication delays $T(t)$, the objective is to design the controllers u_l and u_r that obtain the performance as follows.

1) *Stability:* Guarantee the closed-loop stability of teleoperation system under communication delays, nonlinearities, and various uncertainties.
2) *Position and force tracking:* Let remote robot accurately track the desired motion trajectory $X_l(t)$ sent from the local robot, and accurately track the task-based force trajectory $F_d(t)$ for obtaining the precise contact force F_e. So the tracking errors are requested that $e_{rm} = X_r(t) - [K_{lr,1}X_l(t - T(t)) + K_{lr,2}]$ and $e_{rf} = F_e - F_d$ to be as small as possible, where $K_{lr,1}$ and $K_{lr,2}$ are the scaling factors.
3) *Force feedback:* The force feedback in the local side can accurately reflect the environment in various conditions, i.e., the human operator will not feel any force feedback during free motion, and sense the well-tracked contact force during rigid contact. The good contact behavior [22] is defined in local side to guarantee that the human operator can accurately sense the contact force between the remote robot and environment, which satisfies

$$M_{ld}(X_l)\ddot{X}_l + C_{ld}\left(X_l, \dot{X}_l\right)\dot{X}_l + G_{ld}(X_l) = F_h - K_{lr,f}\hat{F}_e \tag{8}$$

where $\hat{F}_e$ is the force feedback, M_{ld}, C_{ld}, and G_{ld} are the positive virtual contact parameters of local robot, $K_{lr,f}$ is the scaling factor.

III. Hybrid Motion/Force Control for Teleoperated Manipulation

A. Overall Control Framework

Focusing on the uncertain contact of remote robot, a novel teleoperation control framework is proposed and shown in Fig. 2, where the only trajectory transmitted from the local side to the remote side is the position $X_l(t)$ of the local robot. Considering the different structure of local and remote robots, the scaling factors $K_{lr,1}$ and $K_{lr,2}$ are designed after receiving the trajectory $X_l(t - T(t))$ through the communication channel, and the motion trajectory $X_{re} = K_{lr,1}X_l(t - T(t)) + K_{lr,2}$ to be tracked is obtained via a filter to best adapt to the motion range between the local and remote robot. Moreover, since the manipulation tasks may require the precise and secure contact, the task-based force trajectory F_d is set to guarantee the required contact between the remote robot and environment.

Considering that the various conditions, such as free motion and rigid contact exist in the interaction between the remote robot and environment, a unified contact model is designed to be the targeted environment interaction. This model can cover these conditions in one equation by setting different model parameters. Particularly, since there exists the situation that the contact force as well as the stiffness or damping parameters may suddenly change when the remote robot starts the rigid contact condition, which may lead to the misteleoperation of human operator, or violate the state constraints of remote robot. As a result, the remote robot may destroy the objects or environment when the contact force is large enough. To avoid this situation, a two-loop

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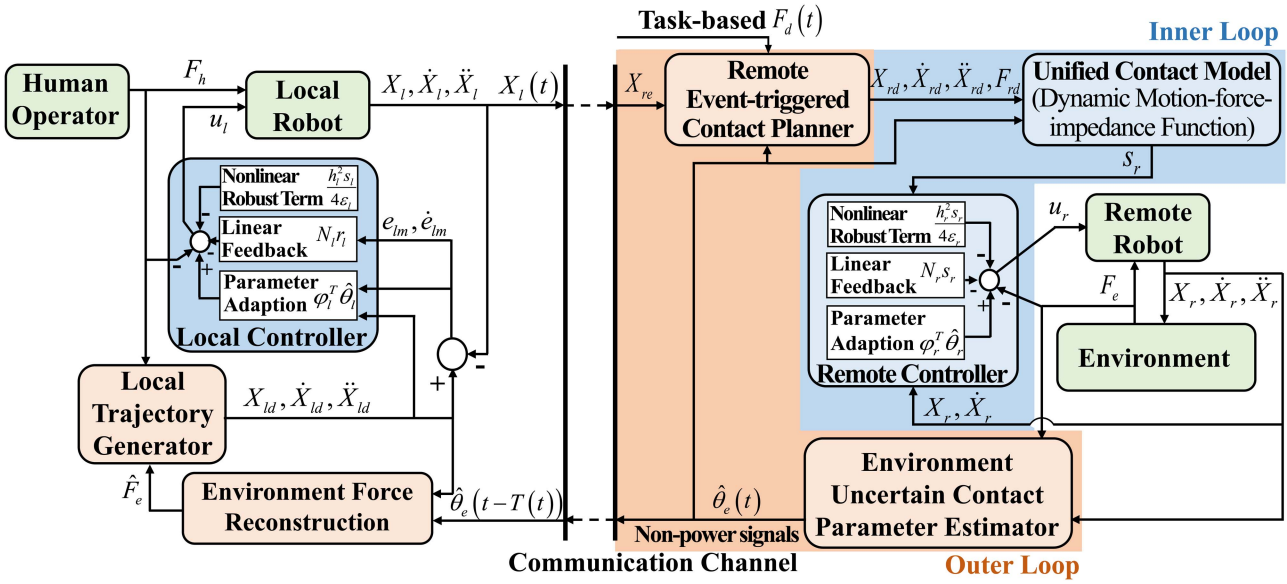


Fig. 2. Novel teleoperation control framework for the hybrid motion/force control of teleoperated manipulation.

control design is implemented. In the outer loop, the remote state-triggered contact planner is designed to plan the desired motion and force trajectories X_{rd} and F_{rd}, which meet the state constraints of remote robot and targeted interaction constraint of unified contact model. Also, these trajectories guarantee the motion to be stable and smooth from various conditions. Subsequently, the remote hybrid motion/force controller is designed in the inner loop, which can achieve the good tracking of desired X_{rd} and F_{rd} so that the system is transiently convergent to the unified contact model. Therefore, the precise and secure contact force is obtained, so that the remote robot is more compliant to the environment, thus reducing the workload of human operator during the teleoperated manipulation.

Since delays inevitably exist in the teleoperation system, the system will be unstable due to the power-cycle (e.g., the position and force signals) in the communication channel. Instead, the signal transmitted in the communication channel is replaced from the environment force F_e to the estimated environment parameter $\hat{\theta}_e$, which effectively eliminates the power-cycle and avoids the traditional passivity stability problem. Moreover, with the transmitted parameter $\hat{\theta}_e$ and matched dynamics of local robot, the environment dynamics is reconstructed as $\hat{F}_e$ and served as force feedback. As a result, the human operator can accurately sense the various conditions in the interaction between the remote robot and environment, especially the compliant contact force guaranteed by hybrid motion/force controller. Afterward, the good contact behavior (8) is achieved and the trajectory X_{ld} can be obtained simultaneously. By the good tracking of local robot achieved via the local controller, the position $X_l(t)$ is obtained, which contains the human operator's intention adjusted via the force feedback. And this position trajectory $X_l(t)$ is transmitted to the remote side, and used to implement the teleoperated manipulation, as described at the beginning of this section.

B. Unified Contact Model and Dynamic Characteristics Analysis

During the teleoperation manipulation, there usually exist the various conditions such as free motion and rigid contact in the interaction between the remote robot and environment. To cover all these conditions in one equation, an unified contact model using a dynamic motion–force–impedance function is designed at first in this section.

In detail, Fig. 1 shows the rigid contact condition between the remote robot and environment. To better study the dynamic characteristics of motion, force, and impendence for the diffident conditions, such as free motion and rigid contact, a dynamic motion–force–impedance function is designed as follows:

$$M_{rd}(\ddot{X}_r - \ddot{X}_{rd}) + C_{rd}(\dot{X}_r - \dot{X}_{rd}) + K_{rd}(X_r - X_{rd}) = -K_{rf}(F_e - F_d) \tag{9}$$

where M_{rd}, C_{rd}, K_{rd}, and K_{rf} are the positive diagonal model parameter matrices, X_{rd} is the desired position trajectory to be tracked by remote robot, which will be discussed later.

Substitute the environment dynamics (7) into (9), then, the interacted relationship between the remote robot and environment can be derived as follows:

$$M_{rd}\ddot{X}_r + (C_{rd} + K_{rf}C_{re})\dot{X}_r + (K_{rd} + K_{rf}K_{re})X_r = M_{rd}\ddot{X}_{rd} + C_{rd}\dot{X}_{rd} + K_{rd}X_{rd} + K_{rf}F_d + K_{rf}K_{re}X_{rc}. \tag{10}$$

When the interacted relationship tends to the steady state, that means $\dot{X}_r = \dot{X}_{rd} = 0$ and $\ddot{X}_r = \ddot{X}_{rd} = 0$, then (10) can be simplified as follows:

$$(K_{rd} + K_{rf}K_{re})X_r = K_{rd}X_{rd} + K_{rf}F_d + K_{rf}K_{re}X_{rc} \tag{11}$$

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and the steady state motion tracking error e_{rms} is described as follows:

$$e_{\text{rms}} = (K_{rf}^{-1}K_{rd} + K_{re})^{-1}[K_{re}(X_{rc} - X_{rd}) + F_d]. \quad (12)$$

Similarly, the steady state of (7) can be obtained as $K_{re}(X_r - X_{rc})$, which can be substituted in (11), then the steady state force tracking error e_{rfs} is described as follows:

$$e_{rfs} = -K_{rf}^{-1}K_{rd}e_{\text{rms}}. \quad (13)$$

Remark 1: Based on (12) and (13), it can be obtained that when the environment stiffness K_{re} is low, the matrices can be selected as $K_{rd} \gg K_{rf}$, then the motion tracking is emphasized and the e_{rms} is relatively small while the e_{rfs} is relatively large. Otherwise, when the environment stiffness K_{re} is high, the matrices can be selected as $K_{rd} \ll K_{rf}$, then the force tracking is emphasized and the e_{rms} is relatively large while the e_{rfs} is relatively small. ■

Moreover, from (10), one can be obtained that

$$\begin{aligned} &M_{rd}(\ddot{X}_r - \ddot{X}_{rd}) + (C_{rd} + K_{rf}C_{re})(\dot{X}_r - \dot{X}_{rd}) \\ &+ (K_{rd} + K_{rf}K_{re})(X_r - X_{rd}) \\ &= -K_{rf}[K_{re}(X_{rd} - X_{rc}) + C_{re}\dot{X}_{rd} - F_d]. \end{aligned} \quad (14)$$

Remark 2: Note that $K_{rf} \neq 0$ during the environment contact condition, based on (14), for any positive M_{rd}, C_{rd}, K_{rd}, and K_{rf}, if the desired trajectories X_{rd} and F_d meet the targeted interaction constraint described as follows:

$$K_{re}(X_{rd} - X_{rc}) + C_{re}\dot{X}_{rd} - F_d = 0 \quad (15)$$

then, the motion tracking error $X_r - X_{rd} \to 0$, and based on (9), the force tracking error $F_e - F_d \to 0$. ■

Based on the above-mentioned analysis, the proposed unified contact model (9) works at the following two parts.

1) As illustrated in Remark 1, by the different selection of model parameters M_{rd}, C_{rd}, K_{rd}, and K_{rf}, the unified contact model (9) combines the dynamic characteristics of motion, force, and impendence in one equation, which appears in the diffident conditions, such as free motion and rigid contact.
2) As illustrated in Remark 2, the unified contact model (9) can be regarded as the targeted environment interaction for remote robot. More specifically, (9) can be used to guide the hybrid motion/force controller design, which helps meet the targeted interaction constraint (15) of remote robot and finally achieve the targeted environment interaction as (9). The detailed controller design will be discussed in the next section.

C. Remote Hybrid motion/force Controller Design

Define the motion tracking error as: $e_{rm} = X_r - X_{rd}$, and the force tracking error as: $e_{rf} = F_e - F_{rd}$, where X_{rd} and F_{rd} are the desired motion and force trajectories, which will be discussed in the next section. So the unified contact model (9) is derived as follows:

$$M_{rd}\ddot{e}_{rm} + C_{rd}\dot{e}_{rm} + K_{rd}e_{rm} = -K_{rf}e_{rf}. \quad (16)$$

Subsequently, inspired by the idea of force control in [23], the following remote hybrid motion/force controller is designed for remote robot, which extends the teleoperation system that can be used for manipulation in contact environment.

First, define the dynamic term as follows:

$$\dot{\omega}_r + B_r\omega_r = E_{rp}e_{rm} + E_{rc}\dot{e}_{rm} + E_{rf}e_{rf} \quad (17)$$

where ω_r is the state of dynamic term, B_r is the positive diagonal matrix, E_{rp}, E_{rc}, and E_{rf} will be designed later.

Then, a sliding surface s_r can be designed as follows:

$$s_r(e_{rm}, \dot{e}_{rm}, \omega_r) = \dot{e}_{rm} + W_{r1}e_{rm} + W_{r2}\omega_r \quad (18)$$

where W_{r1}, W_{r2} is the positive diagonal matrix, and W_{r2} should be nonsingular.

Take the derivative of (18) as $\dot{s}_r = \ddot{e}_{rm} + W_{r1}\dot{e}_{rm} + W_{r2}\dot{\omega}_r$, and substitute it with (17) and (18), one can be obtained that

$$\begin{aligned} &\ddot{e}_{rm} + (W_{r1} + W_{r2}B_rW_{r2}^{-1} + W_{r2}E_{rc})\dot{e}_{rm} \\ &+ (W_{r2}E_{rp} + W_{r2}B_rW_{r2}^{-1}W_{r1})e_{rm} \\ &= -W_{r2}E_{rf}e_{rf} + \dot{s}_r - W_{r2}B_rW_{r2}^{-1}s_r. \end{aligned} \quad (19)$$

Since $W_{r2}B_rW_{r2}^{-1} = B_r$, the E_{rp}, E_{rc}, and E_{rf} in (17) can be designed as follows:

$$\begin{cases} E_{rp} = W_{r2}^{-1}(M_{rd}^{-1}K_{rd} - B_rW_{r1}) \\ E_{rc} = W_{r2}^{-1}(M_{rd}^{-1}C_{rd} - W_{r1} - B_r) \\ E_{rf} = W_{r2}^{-1}M_{rd}^{-1}K_{rf}. \end{cases} \quad (20)$$

Substitute (20) into (19), one can be obtained that

$$M_{rd}\ddot{e}_{rm} + C_{rd}\dot{e}_{rm} + K_{rd}e_{rm} = -K_{rf}e_{rf} + M_{rd}(\dot{s}_r + B_rs_r). \quad (21)$$

It can be obtained that if the sliding surface s_r is convergent, the control objective (16) holds. Therefore, the remote hybrid motion/force controller controller u_r can be designed as follows:

$$u_r = \varphi_r^T(X_r, \dot{X}_r, \dot{X}_{rq}, \ddot{X}_{rq})\hat{\theta}_r - F_e - N_rs_r - \frac{1}{4\varepsilon_r}h_r^2s_r \quad (22)$$

where $\hat{\theta}_r$ is the estimation of parameters in dynamics (2) of remote robot and complies with Property 3, the estimation error $\tilde{\theta}_r = \theta_r - \hat{\theta}_r$, $\dot{X}_{rq} = \dot{X}_{rd} - W_{r1}e_{rm} - W_{r2}\omega_r$, $\ddot{X}_{rq} = \ddot{X}_{rd} - W_{r1}\dot{e}_{rm} - W_{r2}\dot{\omega}_r$, $N_r > 0$, $h_r \geq \|\varphi_r(X_r, \dot{X}_r, \dot{X}_{rq}, \ddot{X}_{rq})\|\|\theta_{r,\max} - \theta_{r,\min}\| + \delta_r$ that complies with Assumption 1.

Specifically, the adaptive law to real time update $\hat{\theta}_r$ in (22) is designed as follows:

$$\dot{\hat{\boldsymbol{\theta}}}_r = \text{Proj}_{\hat{\theta}_r}(\Gamma_r\tau_r) \quad (23)$$

where $\Gamma_r > 0$, $\tau_r = \varphi_r(X_r, \dot{X}_r, \dot{X}_{rq}, \ddot{X}_{rq})s_r$, $\text{Proj}_{\hat{\theta}_r}(\bullet)$ is the projection mapping function defined as follows:

$$\text{Proj}_{\hat{\theta}_i}(\bullet_i) = \begin{cases} 0 & \text{if } \hat{\theta}_i = \theta_{i,\max} \text{and} \bullet_i > 0 \\ 0 & \text{if } \hat{\theta}_i = \theta_{i,\min} \text{and} \bullet_i < 0 \\ \bullet_i & \text{otherwise.} \end{cases} \quad (24)$$

Apart from the property in (5), the following desirable properties is guaranteed for any adaption laws when the projection mapping

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in (24) is used

$$\tilde{\theta}_i^T[\Gamma_i^{-1}\text{Proj}_{\hat{\theta}_i}(\Gamma_i\tau_i)-\tau_i]\leq 0,\forall\tau_i. \tag{25}$$

Theorem 1: Considering the dynamics (2) of remote robot subject to the unified contact model (16), based on the bounded desired motion and force trajectories X_{rd} and F_{rd}, and measured F_e, by applying the remote hybrid motion/force controller (22) and the adaptive law (23), then the following properties can be obtained.

1) All of the signals in the remote subsystem can be guaranteed to be bounded, and the remote subsystem is stable. Meanwhile, the Lyapunov function $V_{r1}=\frac{1}{2}s_r^T M_r s_r$ including the sliding surface s_r is bounded by

$$V_{r1}(t)\leq \exp\left(-\lambda_{rv}t\right)V_{r1}(0)+\frac{\varepsilon_r}{\lambda_{rv}}\left[1-\exp\left(-\lambda_{rv}t\right)\right] \tag{26}$$

where $\lambda_{rv}=\frac{2\lambda_{r,\min}(N_r)}{\eta_{r2}}$, η_{r2} complies with Property 1.

2) If there exist the constant parametric uncertainties only after the finite time t_0 (i.e., $\dot{\tilde{d}}_r=0,\forall t>t_0$), $s_r\to 0$ when $t\to\infty$, and the unified contact model (16) satisfies

$$M_{rd}\ddot{e}_{rm}+C_{rd}\dot{e}_{rm}+K_{rd}e_{rm}+K_{rf}e_{rf}\to 0 \tag{27}$$

and $e_{rm}\to 0$, $e_{rf}\to 0$ when $t\to\infty$.

D. Remote Event-Triggered Contact Planner Design With Model Predictive Control (MPC)-Based Contact Optimization

Considering that the remote robot has the different structure than that of the local robot, the trajectory to be planned should be modified to match the work space between the local and remote robots, so the position trajectory from the local robot to the remote robot are redefined as follows:

$$X_{re}=K_{lr,1}X_l\left(t-T(t)\right)+K_{lr,2} \tag{28}$$

and a filter $V_{rf}(s)=1(1+\tau_{rf}s)^2$ is then used to calculate the (28) and obtain the velocity $\dot{X}_{re}$ and acceleration $\ddot{X}_{re}$, respectively.

Particularly, in the teleoperation process, there exists the situation that the remote robot changes the condition from the free motion to the contact one when implementing the manipulation tasks. In this situation, the model parameters M_{rd}, C_{rd}, K_{rd}, and K_{rf} in (9) as well as the environment parameter K_{re} and C_{re} in (7) may suddenly change. So the human operator may misteleoperate the remote robot, or even destroy the objects or environment if the contact force is large enough. All these sudden change and misteleoperation will violate the targeted interaction constraint (15) and state constraints of remote robot [24], [25] (e.g., velocity, acceleration, contact force, etc.). As a result, the targeted environment interaction (9) may not be achieved even if the hybrid motion/force controller is well designed.

Therefore, a two-loop control design is implemented for the remote subsystem, where the hybrid motion/force controller (22) is used as the inner loop, and a remote state-triggered contact planner is designed as the outer loop, which is used to plan the acceptably desired motion and force trajectories that meet the targeted interaction constraint (15) and state constraints of remote robot.

Define a state-triggered function as follows:

$$V_{\text{etc}}=F_e^T\Lambda F_e \tag{29}$$

where Λ is the positive diagonal matrix, then the state-triggered strategy can be described as follows.

State 1: When $V_{\text{etc}}\leq\|\bar{F}_{\max}\|^2$, where $\|\bar{F}_{\max}\|>\|F_d\|$ is the maximum acceptable contact force exerted on the environment, which means that the manipulation task is normally teleoperated, so let $X_{rd}=X_{re}$ and $F_{rd}=F_d$. In such situation, the remote robot is directly teleoperated by the human operator via X_{re}, and the remote hybrid motion/force controller (22) in the inner loop can guarantee the transient convergence of system to the targeted environment interaction as (9).

State 2: When $V_{\text{etc}}>\|\bar{F}_{\max}\|^2$, which means the sudden change of condition or misteleoperation occurs, thus violating the targeted interaction constraint (15) and state constraints of remote robot. In such situation, the remote robot is not well interacted with the environment, so it stops being teleoperated by the human operator, instead, the new trajectory X_{rcf}, which satisfies: $X_{rcf}\geq\hat{K}_{re}^{-1}F_d+X_{rc}$, is set as the desired trajectory for the tracking of remote robot to resume the contact force to be F_d as quickly as possible. Subsequently, an MPC-based contact optimization is implemented to obtain the desired X_{rd} and F_{rd} that meet the targeted interaction constraint (15) and state constraints of remote robot. The optimization problem can be described as follows:

$$\begin{aligned}\min_{X_{rd},F_{rd}} J=&\int_{t_0}^{t_0+t_r}\left[\frac{1}{2}\left(X_{rd}-X_{rcf}\right)^T P\left(X_{rd}-X_{rcf}\right)\right.\\&\left.+\frac{1}{2}\left(F_{rd}-F_d\right)^T Q\left(F_{rd}-F_d\right)\right]dt\\ \text{s.t.}&\begin{cases}\hat{K}_{re}\left(X_{rd}-X_{rcf}\right)+\hat{C}_{re}\dot{X}_{rd}=F_{rd}\\ X_{rd}\left(t_0\right)=X_r\left(t_0\right)\\ \dot{X}_{\min}\leq\dot{X}_{rd}\leq\dot{X}_{\max}\\ \ddot{X}_{\min}\leq\ddot{X}_{rd}\leq\ddot{X}_{\max}\\ 0\leq\|F_{rd}\|\leq\|\bar{F}_{\max}\|\end{cases}\end{aligned} \tag{30}$$

where t_0 and t_r are the initial time and predict duration of each planning sampling period, P and Q are the weighting matrices, $\dot{X}_{rd}$ and $\ddot{X}_{rd}$ are the planned velocity and acceleration of the X_{rd}, $\dot{X}_{\max}$, and $\dot{X}_{\min}$ are the maximum and minimum value of the allowable velocity, $\ddot{X}_{\max}$ and $\ddot{X}_{\min}$ are the maximum and minimum value of the allowable acceleration. The initial value of $X_{rd}(t_0)$ at $t=t_0$ during the planning will be set as the real state $X_r(t_0)$ of remote robot, which is to guarantee the smooth and continuous of planned trajectory when the state changes.

To guarantee the real time of optimization problem (30), it is derived as a quadratic programming problem, which is described as follows:

$$\begin{aligned}&\min\frac{1}{2}f_{rs}^T H\left(t_0\right)f_{rs}+f_{rs}^T\nabla\left(t_0\right)\\ &\text{s.t.}\begin{cases}lbW\left(t_0\right)\leq Wf_{rs}\leq ubW\left(t_0\right)\\ lb\left(t_0\right)\leq f_{rs}\leq lb\left(t_0\right)\end{cases}\end{aligned} \tag{31}$$

where f_{rs} is a sequence of F_{rd} on time, H is the Hessian matrix related to P and Q in (30), $\nabla(t_0)$ is the gradient vector related to

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$X_r(t_0)$, P and environment constraints, $lbW(t_0)$ and $ubW(t_0)$ are obtained from $\dot{X}_{\min}$, $\dot{X}_{\max}$, $\ddot{X}_{\min}$, and $\ddot{X}_{\max}$, $lb(t_0)$ and $ub(t_0)$ are obtained from 0 and $\|\bar{F}_{\max}\|$.

Note that if the planning of State 2 is finished, the remote robot is recovered from the large contact force, and the human operator can directly sense the decrease of the contact force via the force feedback $\hat{F}_e$. Since the replanned trajectories meet the targeted interaction constraint (15) and state constraints of remote robot, the State 1 continues to work afterward, where the real state X_r of remote robot can be quickly convergent from X_{rcf} to the newly desired teleoperated motion trajectory X_{re}, which is achieved by the remote hybrid motion/force controller (22) in the inner loop.

E. Environment Dynamics Estimation and Reconstruction

Since communication delays inevitably exist in teleoperation, the system will be unstable due to the power-cycle in the communication channel. To avoid this problem, the environment parameter in (7) is required to accomplish the nonpower signal transmission, which replace the direct force transmission so that the power-cycle is no longer existed. Therefore, to guarantee the system stability under delays and provide the good force feedback, the environment parameter is estimated and transmitted to reconstruct the environment dynamics in this section. The environment dynamics (7) can be written in the regressor form as follows:

$$F_e = \varphi_e^T \theta_e \tag{32}$$

where $\varphi_e = [X_r - X_{rcf}, \dot{X}_r]^T$, $\theta_e = [K_{re}, C_{re}]^T$, X_{rcf} is the position of remote robot recorded when the remote robot first contact the environment.

To obtain the estimated environment parameter $\hat{\theta}_e$, the environment uncertain contact parameter estimator is designed in the remote side using the recursive least squares estimation (RLSE), and the cost function is designed as follows:

$$\digamma(t, \hat{\theta}_e) = \int_0^t \| A(t,\alpha)[F_e(\alpha) - \varphi_e^T(\alpha)\hat{\theta}_e] \|^2 d\alpha \tag{33}$$

where $A(t,\alpha)$ is the weighting matrix.

Subsequently, the RLSE adaptive law to obtain the optimal $\hat{\theta}_e$ can be derived as follows:

$$\dot{\hat{\boldsymbol{\theta}}}_e = \lambda_e \varphi_e \varepsilon_e \tag{34}$$

where $\lambda_e > 0$, and $\varepsilon_e = F_e - \varphi_e^T \hat{\theta}_e$ is the uniformly bounded estimation error.

Therefore, based on the RLSE, the accurate estimated environment parameter $\hat{\theta}_e(t)$ is obtained and transmitted to the local side via the communication channel. The environment force is then reconstructed in the local side to provide the human operator with force feedback, and the reconstructed environment force can be described as follows:

$$\hat{F}_e = \varphi_{el}^T \hat{\theta}_e (t - T(t)) \tag{35}$$

where $\varphi_{el} = [K_{lr,1}(X_{ld} - X_{lcf}) + K_{lr,2}, K_{lr,1}\dot{X}_{ld}]^T$, X_{lcf} is the virtual contact position of local robot recorded when the remote robot first contact the environment, $\hat{\theta}_e = [\hat{K}_{re}, \hat{C}_{re}]^T$.

F. Local Trajectory Generator and Controller Design

Based on the reconstructed $\hat{F}_e$ in the previous section, and the good contact behavior (8), then the desired motion trajectories X_{ld}, $\dot{X}_{ld}$, and $\ddot{X}_{ld}$ to be tracked for the local robot can be generated by the local trajectory generator as follows:

$$M_{ld}(X_{ld})\ddot{X}_{ld} + C_{ld}\left(X_{ld}, \dot{X}_{ld}\right)\dot{X}_{ld} + G_{ld}(X_{ld}) = F_h - K_{lr,f}\hat{F}_e. \tag{36}$$

Note that the human operator can accurately sense the reconstructed $\hat{F}_e$ as force feedback via (36), the generated desired motion trajectory X_{ld} contains the human operator's intention to guide the manipulation for the remote robot.

Subsequently, the local controller is designed to achieve the good tracking of X_{ld} for local robot.

First, define a switching-function-like quantity as follows:

$$s_l = \dot{e}_{lm} + B_l e_{lm} \tag{37}$$

where $e_{lm} = X_l - X_{ld}$ is the motion tracking error of the local robot, B_l is the positive diagonal matrix.

Then, the local controller u_l can be designed as follows:

$$u_l = \varphi_l^T(X_l, \dot{X}_l, \dot{X}_{lq}, \ddot{X}_{lq})\hat{\theta}_l - F_h - N_l s_l - \frac{1}{4\varepsilon_l} h_l^2 s_l \tag{38}$$

where $\hat{\theta}_l$ is the estimation of parameters in dynamics (1) of local robot and complies with Property 3, the estimation error $\tilde{\theta}_l = \theta_l - \hat{\theta}_l$, $\dot{X}_{lq} = \dot{X}_{ld} - B_l e_{lm}$, $\ddot{X}_{lq} = \ddot{X}_{ld} - B_l \dot{e}_{lm}$, $N_l > 0$, $h_l \geq \|\varphi_l(X_l, \dot{X}_l, \dot{X}_{lq}, \ddot{X}_{lq})\|\|\theta_{l,\max} - \theta_{l,\min}\| + \delta_l$ that complies with Assumption 1.

Specifically, the adaptive law to real time update $\hat{\theta}_l$ in (38) is designed as follows:

$$\dot{\hat{\boldsymbol{\theta}}}_l = \mathrm{Proj}_{\hat{\theta}_l}(\Gamma_l \tau_l) \tag{39}$$

where $\Gamma_l > 0, \tau_l = \varphi_l(X_l, \dot{X}_l, \dot{X}_{ld}, \ddot{X}_{ld})s_l$, $\mathrm{Proj}_{\hat{\theta}_i}(\bullet)$ is the projection mapping function defined as (24).

Theorem 2: Considering the dynamics (1) of local robot, based on the bounded motion trajectory X_{ld} and measured F_h, by applying the local controller (38) and the adaptive law (39), then the following properties can be obtained.

1) All of the signals in the local subsystem can be guaranteed to be bounded, and the local subsystem is stable. Meanwhile, the Lyapunov function $V_{l1} = \frac{1}{2} s_l^T M_l s_l$ including the sliding surface s_l is bounded by

$$V_{l1}(t) \leq \exp(-\lambda_{lv} t) V_{l1}(0) + \frac{\varepsilon_l}{\lambda_{lv}} [1 - \exp(-\lambda_{lv} t)] \tag{40}$$

where $\lambda_{lv} = \frac{2\lambda_{l,\min}(N_l)}{\eta_{l2}}$, η_{l2} complies with Property 1.

2) If there exist the constant parametric uncertainties only after the finite time t_0 (i.e., $\tilde{d}_l = 0, \forall t > t_0$), $s_l \to 0$ when $t \to \infty$, then $e_{lm} \to 0$ when $t \to \infty$.

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G. Performance Analysis

This section summarizes how this framework achieves the objective described in the problem formulation.

Remark 3: With the RLSE-based adaptation for environment dynamics (7), the estimated environment parameter $\hat{\theta}_e$ is used to replace the environment dynamics F_e, and transmitted in the communication channel. That means, the power-cycle formed by the forward position and feedback force signals no longer exists, and the signals transmitted in the communication channel can be guaranteed to be bounded. Therefore, the traditional passivity stability problem is avoided, and the stability of the delayed teleoperation system can be guaranteed by the separate design of local and remote controllers. In detail, with the bounded $\theta_e(t-T(t))$, the reconstructed environment force $\hat{F}_e$ is bounded, and by the proper selection of M_{ld}, C_{ld}, and G_{ld} in (36), the desired motion trajectories X_{ld}, $\dot{X}_{ld}$, and $\ddot{X}_{ld}$ can be guaranteed to be bounded. Since the local controller (38) can guarantee the good tracking of local robot to X_{ld}, so X_l is bounded. By the transmission of X_l through communication channel and filtering, X_{re} is bounded. With the bounded X_{re} and F_d, the remote state-triggered contact planner can guarantee the desired X_{rd} and F_{rd} to be bounded. And the remote hybrid motion/force controller (22) can guarantee the good tracking of remote robot to X_{rd} and F_{rd}, so X_r and F_e are bounded. In summary, all of the signals in the teleoperation system are bounded under communication delays, nonlinearities and various uncertainties, and the stability of system is guaranteed. ■

Remark 4: Since the filter V_{rf} is stable, X_{re} is convergent to the matched $K_{lr,1}X_l(t-T(t))+K_{lr,2}$, and the remote state-triggered contact planner guarantees the convergence of $X_{rd}\to X_{re}$. Therefore, by the good tracking of $X_r\to X_{rd}$ via the remote hybrid motion/force controller (22), the good position tracking of $X_r\to K_{lr,1}X_l(t-T(t))+K_{lr,2}$ is achieved. ■

Remark 5: Since the remote state-triggered contact planner guarantees the convergence of $F_{rd}\to F_d$, and the remote hybrid motion/force controller (22) as well as the unified contact model (9) guarantee the good tracking that $F_e\to F_{rd}$, so the good force tracking of $F_e\to F_d$ is achieved. Therefore, the precise and secure contact force is theoretically guaranteed without concentrated and laborious operation, which effectively reduces the workload of human operator. ■

Remark 6: With the accurate estimated environment parameter $\hat{\theta}_e\to\theta_e$ obtained via the RLSE-based environment uncertain contact parameter estimator, and the good position tracking that $X_r(t)\to K_{lr,1}X_{ld}(t-T(t))+K_{lr,2}$, as well as the transmitted $\hat{\theta}_e(t-T(t))\to\hat{\theta}_e(t)$, then the good force feedback $\hat{F}_e\to F_e$ is achieved. Moreover, by the proper selection of M_{ld}, C_{ld}, and G_{ld} in (36), the good contact behavior is achieved, which better improves the ability of human operator to guide the manipulation of remote robot by planning the human-enforced desired motion trajectories X_{ld}, $\dot{X}_{ld}$, and $\ddot{X}_{ld}$. ■

Remark 7: With the stability achieved in Remark 3, the good position and force tracking achieved in Remark 4 and 5, the good force feedback achieved in Remark 6, then the teleoperation system simultaneously achieves the stability and good transparency under delays, nonlinearities, various uncertainties and uncertain contract. Moreover, since the dynamic characteristics of unified contact model (9) are achieved in Remark 1, and the targeted interaction constraint (15) is satisfied via the remote hybrid motion/force controller (22) in Remark 2 so as to achieve the targeted environment interaction as (9), as a result, the good contact of remote robot is obtained in unknown environment during the teleoperated manipulation. ■

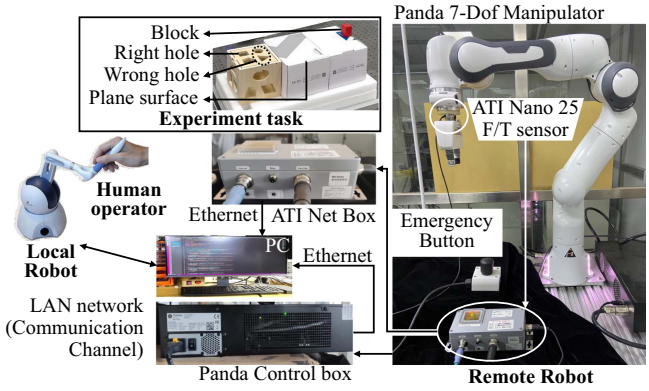


Fig. 3. Real experiment platform.

IV. Experiment

A. Experiment Setup

As shown in Fig. 3, a real experiment platform is used for the teleoperated manipulation in unknown environment with uncertain contact. A Phantom Omni haptic device (three-Dof in force feedback) is used as the local robot with dynamics (1), and a Panda seven-Dof manipulator is used as the remote robot with dynamics (2). Particularly, an ATI Nano 25 F/T sensor is mounted on the end effector of remote robot to obtain the contact force between the remote robot and environment. In the local side, the Phantom Omni is connected to a PC (Intel Core i7-8750H @ 2.20 GHz, running with Windows 10) via Ethernet. And in the remote side, the Panda manipulator and ATI Nano 25 F/T sensor are connected to the Panda control Box and ATI Net Box, which are communicated with a PC (Intel Core i5-12400F @ 2.50 GHz, running with Ubuntu 18.04 real-time kernal, Linux 5.4.19-rt14 under Robot Operating System) via Ethernet. The local robot and remote robot are communicated via LAN network by publishing or subscribing the nodes defined in the robot operating system. The sampling period of the system is 0.001 s.

Since the local and remote robots have the different structure, the scaling factors in (28) are defined as $K_{lr,1}=\text{diag}\{1.5,1.5,1.5\}$ and $K_{lr,2}=[0.5237,0,0.4229]^T$. Besides, since the remote robot has the redundant Dof, its Jacobian matrix $J_r(q_r)$ defined in Section II-A is singular, where $J_r^\dagger=(J_r^TJ_r)^{-1}J_r^T$ is the pseudoinverse of J_r.

To verify the effectiveness of proposed teleoperation control framework in achieving the teleoperated manipulation under the uncertain contact in unknown environments, as well as some inevitable issues, such as nonlinearities, various uncertainties,

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constraints, and communication delays, an experiment of grabbing and placing the block is designed. In detail, the experiment mainly consists of three periods, namely, in Period I, the human operator is asked to teleoperate the remote robot to grab the block; in Period II, the human operator is asked to teleoperate the remote robot to hold the block and moves it on a plane surface, the desired F_d in this period is defined as $F_d = [0, 0, -1]^T$N that the block will move with constant contact force; in Period III, the human operator is asked to teleoperate the remote robot to place the block into the right hole, the desired F_d in this period is defined as $F_d = [0, 0, 0]^T$N that the block will not contact the area near the hole.

In the experiment, the initial states of local and remote robots are $X_l(0) = [-0.0881, 0, -0.0655]^T$m and $X_r(0) = [0.3916, 0, 0.3247]^T$m, respectively. The time-varying delays are simulated as 0.27 ± 0.05 s.

For the fair comparison, two comparative teleoperation control frameworks are implemented as follows:

T1: The wave variables-based framework with direct environment force transmission [14], where the parameter of wave impendence is designed as 5.

T2: The proposed framework with estimated nonpower environment parameter transmission. The initial value of λ_e in (34) is set as $\lambda_e(0) = \text{diag}\{800, 800\}$ in each direction of environment dynamics estimation; $\tau_{rf} = 0.015$; In the local subsystem, the virtual contact parameters are set as $M_{ld} = \text{diag}\{4, 4, 2.5\}$, $C_{ld} = \text{diag}\{0, 0, 0\}$, $G_{ld} = [0, 0, 19.6]^T$; $K_{lr,f} = \text{diag}\{1, 1, 1\}$; $B_l = \text{diag}\{1.5, 1.5.1, 5\}$, the hybrid value of $N_l + \frac{1}{4\varepsilon_l} h_l^2$ is set as $\text{diag}\{9, 35, 15\}$, Γ_l is set as $\text{diag}\{4, 4, 25, 25, 2, 2\}$ in (39).

Specifically, to verify that the proposed unified contact model and two-loop control design can achieve the better compliance of remote robot with precise and secure contact force, two different remote subsystems are compared as follows.

C1: The remote subsystem with only the motion control, which has the same structure with the local subsystem designed in Section III-F, and the traditional damper–spring system is used as the environment dynamics.

C2: The remote subsystem with proposed two-loop control design, which includes the MPC-based contact optimization and remote hybrid motion/force controller (22), and the proposed unified contact model (9) is added with the environment dynamics (7). For the two-loop control design, the parameters in the inner loop are set as: $B_r = \text{diag}\{2, 2, 2\}$, $W_{r1} = \text{diag}\{1, 1, 1\}$, $W_{r2} = \text{diag}\{10, 10, 10\}$, the hybrid value of $N_r + \frac{1}{4\varepsilon_l} h_r^2$ is $\text{diag}\{50, 50, 50\}$, $\Gamma_r = \text{diag}\{10, 10, 125, 125, 2, 2, 25\}$; the parameters in the outer loop are set as: $\Lambda = \text{diag}\{1, 1, 1\}$, $P = \text{diag}\{1, 1, 1\}$, $Q = \text{diag}\{0.002, 0.002, 0.002\}$, $\dot{X}_{\max} = 0.1$m/s, $\ddot{X}_{\max} = 2\text{m}^2/\text{s}$, $\|\bar{F}_{\max}\| = 4.5$ N, the contact optimization is real time solved through qpOASES [26], [27]. For the unified contact model, the model parameters are set as: $M_{rd} = \text{diag}\{1, 1, 1\}$, $K_{rd} = \text{diag}\{600, 600, 600\}$ in free motion while $K_{rd} = \text{diag}\{600, 600, 1\}$ in rigid contact, $C_{rd} = 2\sqrt{K_{rd} M_{rd}}$, $K_{rf} = \text{diag}\{0, 0, 0\}$ in free motion while $K_{rf} = \text{diag}\{0, 0, 1\}$ in rigid contact.

Therefore, the comparative experiment can be set as follows.

Set 1: T1+C1, i.e., the remote robot works with motion control under the traditional teleoperation control framework.

Set 2: T2+C1, i.e., the remote robot works with motion control under the proposed teleoperation control framework.

Set 3: T2+C2, i.e., the remote robot works with contact optimization and remote hybrid motion/force control under the proposed teleoperation control framework.

B. Experiment Results

The experimental results of Set 1 are shown in Fig. 4–Set 1. Since all of the signals are bounded, the system is stable under communication delays. However, due to the wave reflection in T1, the system has the poor transparency performance, which results in the large oscillation in position tracking [shown in Fig. 4(a)–Set 1] and wrong force feedback that will mislead the human operator [shown in Fig. 4(c)–Set 1]. As a result, the human operator suffers from the large psychological burden and increases the workload when implementing the teleoperated manipulation. Specifically, in Period II, the human operator can hardly sense whether the remote robot contacts the plane surface or not, and the poor position tracking results in the bad guidance of human operator, therefore, the remote robot cannot compliantly move on the plane surface, as shown in the blue curve of Fig. 4(b)–Set 1. Finally, due to the poor position tracking, the remote robot fails to place the block into the right hole in Period III.

The experimental results of Set 2 are shown in Fig. 4–Set 2. Similarly, all of the signals are bounded, so the system with proposed T2 is stable under communication delays. Since the wave reflection no longer exists in T2, the good position tracking is obtained via C1, as shown in Fig. 4(a)–Set 2, where the root-mean-square error (RMSE) of e_{rm} in *X*, *Y*, and *Z* axes are 2, 1.7, and 1.7 mm, respectively. Moreover, with the design of $K_{lr,1}$ and $K_{lr,2}$, the motion trajectory X_r in the remote side has the same shape, direction and proportional size with X_l in the local side, as shown in Fig. 4(b)–Set 2, which means the work spaces of local and remote robots are well matched in T2. As a result, the human operator successfully teleoperates the remote robot to grab the block in Period I.

However, since Set 2 only implements the motion controller (C1), the remote robot oscillates when moving along the plane surface at t=20–21 s and 28–32 s, this is because the interactions at t=20–21 s and 28–32 s violate the targeted interaction constraint (15), but the only C1 cannot achieve the targeted environment interaction as (9), so that the remote robot is not compliant to the plane surface, which performs a bad interaction performance in Period II. With the good force feedback achieved in Section III-E for T2, as shown in Fig. 4(c)–Set 2, the human operator can accurately sense the contact force, but the bad interaction performance leads to the continuously and fast changing contact force in Period II, which is up to 8.72 N at t=20.2 s. Such large contact force feedback may be destructive, so the human operator may pay much attention to avoid the excessive instantaneous contact force. Subsequently, in Period III, the accurate force feedback let the human operator be aware

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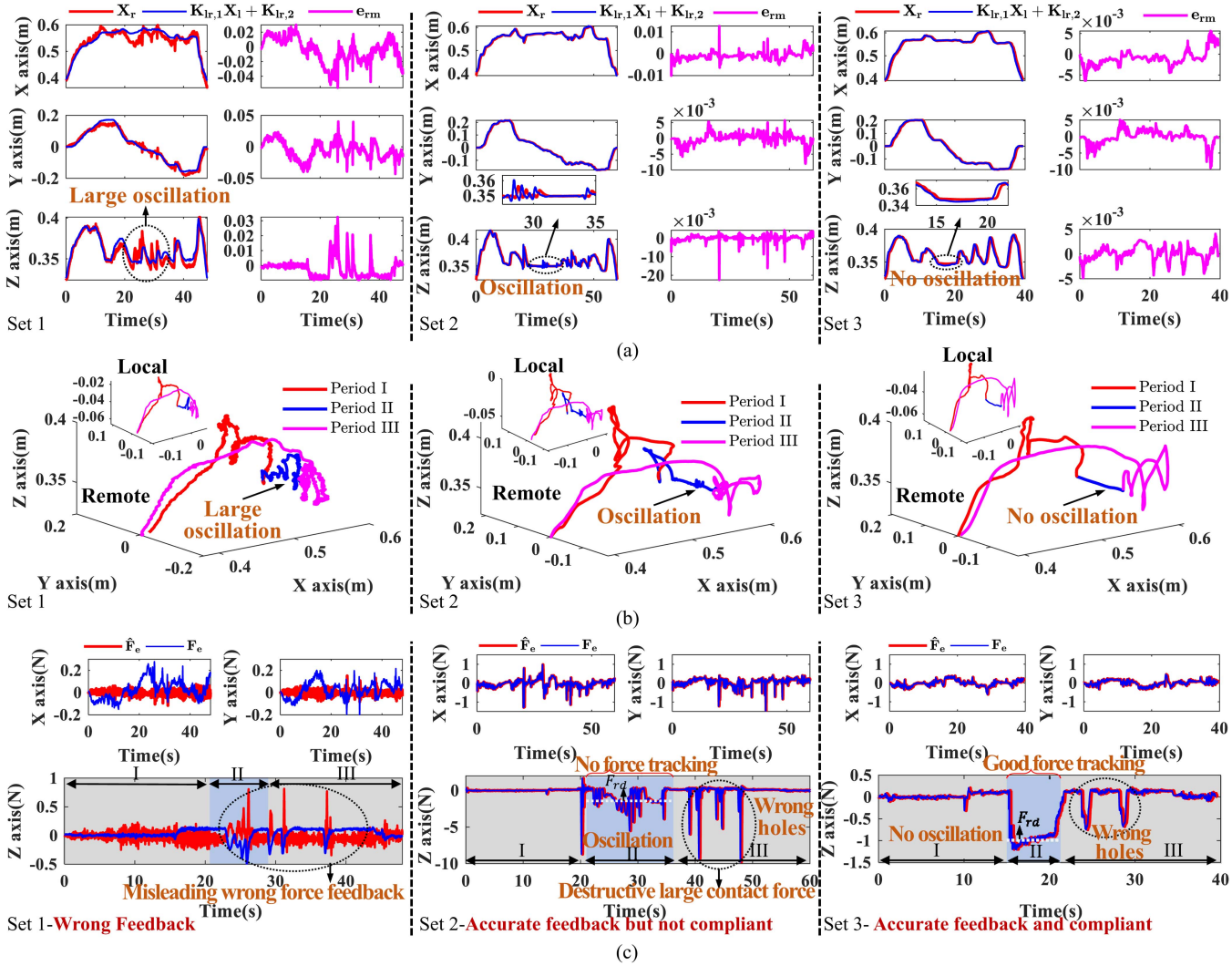


Fig. 4. Experiment results. (a) Position tracking with tracking error. (b) Motion trajectories in local and remote sides. (c) Force feedback.

of the block contacting the wrong holes at t=38–49 s. But due to the bad interaction performance in Set 2, the oscillation happens again in this period. To make things worse, the peak value of the contact force in the Z-axis reaches 9.75 N at t=47.9 s, which may cause damage to the object or the environment. Although the remote robot can place the block into the right hole in Set 2, the whole process in not compliant, and the workload of human operator still increases, which requires the continuous high concentration to find the appropriate contact position between the remote robot and environment.

The experimental results of Set 3 are shown in Fig. 4–Set 3∼ Fig. 6. With the proposed T2, all of the signals are still bounded, and the system is stable under communication delays. Different from Set 2, since Set 3 implements the proposed hybrid motion/force controller (C2) and unified contact model (9), the oscillation no longer exists in Period II, this is because the targeted interaction constraint (15) can be guaranteed via C2, so that the remote robot can be teleoperated to move in the plane surface with constant F_{rd} at t=15–21 s, as shown in Fig. 4(c)–Set 3 and Fig. 5(b), which performs a good and compliant interaction performance in Period II. As a result, the good position tracking is also achieved via C2, as shown in Fig. 4(a)–Set 3, and the compliant force tracking avoids the oscillation, thus leading to the smaller e_{rm} when compared to

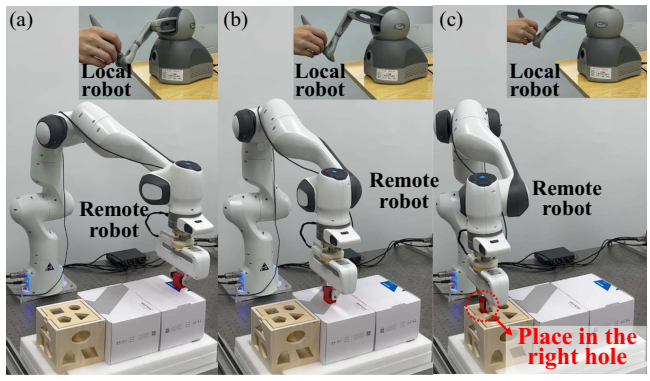


Fig. 5. Teleoperated manipulation process of remote robot in Set 3. (a) Period I. (b) Period II. (c) Period III.

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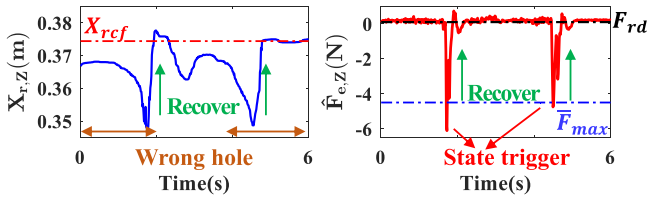


Fig. 6. Motion and contact force feedback of remote robot in Set 3–*Z*-axis when the state-trigger occurs.

that in Set 2, where the RMSE of e_{rm} in *X*, *Y*, and *Z* axes are 1.9, 1.6, and 1.5 mm, respectively. Since the good force feedback is achieved in T2, as shown in Fig. 4(c)–Set 3, the human operator can accurately sense the compliant and constant contact force feedback guaranteed by C2. Therefore, the human operator no longer needs to carefully find the appropriate contact position, and the large and destructive contact force can be avoided, which effectively reduce the workload of human operator, and the time spent is reduced from 13 to 6 s. Subsequently, the human operator begins the block placing task in Period III, as shown in Fig. 5(c). Different from Set 2, since C2 avoids the oscillation and large contact force, the contact force F_e no longer changes sharply. Instead, the peak value of contact force in *Z*-axis is controlled to be only 0.75 N during the contact of wrong holes, and convergently recovers to the desired F_{rd} designed for this period. Finally, the block is placed to the right hole at t=34 s, thus quickly completing the whole task with reduced workload of human operator.

Particularly, to further verify the effectiveness of proposed MPC-based contact optimization in C2, an additional experiment as Period III is tested for Set 3, and the results are shown in Fig. 6. At about t=1.7 s and t=4.5 s, the contact with two wrong holes generates the contact force F_e larger than $\bar{F}_{\max}$, which violates the targeted interaction constraint (15) and triggers the contact optimization. Subsequently, the motion trajectory X_{rcf} is quickly replanned via MPC, and tracked by the remote robot to quickly recover the contact force that converges to F_{rd}. Moreover, the human operator can sense a quick reverse increase of force feedback when the state triggered, this is resulted from the quick recovery of contact force, and the human operator can be quickly aware of the misteleoperation (too large contract force when contacting the wrong hole). Therefore, with the design of MPC-based contact optimization in C2 as an outer loop, the remote robot can quickly recovers from the unexpected contact, which effectively protects the object from damage and further reduces the workload of human operator.

Based on the above-mentioned analysis, the effectiveness of the proposed teleoperation control framework is verified to guarantee the stability of delayed system, achieve the good position and force tacking of remote robot, and provide the good force feedback while reduce the workload for human operator under time-varying delays, nonlinearities, various uncertainties, and constraints. In addition, the proposed framework allows the teleoperated manipulation of remote robot in a more complicated unknown environment where the robot is more compliant to the uncertain contract.

V. Conclusion

In this article, a novel teleoperation control framework is proposed for teleoperated manipulation under the uncertain contact in unknown environment. A unified contact model is established to define the various conditions that may occur during the teleoperation. This model can also be the targeted environment interaction to guide the design of remote hybrid motion/force controller under nonlinearities and various uncertainties, thus guaranteeing system stability and transient convergence to the unified contact model. By the design and adoption of unified contact model and hybrid motion/force controller, the precise and secure contact force is guaranteed without the concentrated and laborious operation of human operator, which effectively reduces the workload. To better guarantee the system security, the MPC-based contact optimization is designed and served as the outer loop of hybrid motion/force controller, thus avoiding the violation of state constraints and targeted interaction constraints during the teleoperation. Moreover, by the RLSE-based estimation and transmission of environment parameter, the environment dynamics is reconstructed in the local side and served as force feedback, which can accurately reflect the environment in various conditions, thus improving the ability of human operator to guide the manipulation of remote robot by planning the human-enforced desired motion trajectories. Since the estimated parameter is replaced to be transmitted in the communication channel, the passivity stability problem is directly avoided. Therefore, the stability and good transparency under delays are guaranteed by the separate design of local and remote controllers. The comparative experiments are implemented, and the results verify the effectiveness of the proposed framework.

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